OEIS A369871: Prescribed Least Prime Factors and Arbitrarily Many Primitive Triangles

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Abstract

For every prime p and integer $K \geq 2$, infinitely many positive integers with least prime factor p are short legs of at least K distinct primitive Pythagorean triangles. This strengthens the infinitude assertion for A369871. The case of two triangles already has a proof by Aditya P Bhattad recorded in OEIS; no priority is claimed.

1 Definition and statement

OEIS A024411 lists, in increasing order, the positive integers that are the shorter leg of more than one primitive Pythagorean triangle. A369871 records their least prime factors. A primitive triangle has positive integer sides with greatest common divisor one.

Theorem 1. *For every prime p and integer $K \geq 2$, there are infinitely many distinct integers m whose least prime factor is p and which are the shorter leg of at least K distinct primitive Pythagorean triangles. Consequently every prime occurs infinitely often in A369871.*

2 Odd primes

Choose a squarefree odd integer A with least prime factor p, having s distinct prime factors, where $2^s \geq K$. Such an A is obtained by multiplying p by $s-1$ distinct larger primes. For each prime $r > (1+\sqrt{2})A$, set $m = Ar$. For each positive divisor d of A put $e = Ar/d$. Then d, e are coprime and odd, and $e/d = Ar/d^2 \geq r/A > 1+\sqrt{2}$. Define

$$b_d = \frac{e^2 - d^2}{2}, \qquad c_d = \frac{e^2 + d^2}{2}.$$

These are integers satisfying $m^2 + b_d^2 = c_d^2$. Moreover $b_d > de = m$, since $t^2 - 1 > 2t$ when $t = e/d > 1+\sqrt{2}$. If an odd prime divides both de and b_d, it divides either d or e, and the formula for $2b_d$ then forces it to divide the other, a contradiction. Also de is odd. Thus $\gcd(m, b_d) = 1$ and the triangle is primitive. As d increases through the divisors of A, the expression $((Ar/d)^2 - d^2)/2$ strictly decreases. Hence the 2^s triangles are distinct. The least prime factor of m is p, and infinitely many choices of r give distinct m.

3 The prime two

Choose an odd squarefree integer A with s prime factors and $2^s \geq K$. For each prime $r > 2(1+\sqrt{2})A$, set $m = 4Ar$. For every divisor d of A put $u = 2d$ and $v = Ar/d$. These are coprime

integers of opposite parity, and

$$2uv = m, \qquad \frac{v}{u} = \frac{Ar}{2d^2} \geq \frac{r}{2A} > 1 + \sqrt{2}.$$

The sides m, $b_d = v^2 - u^2$, $c_d = v^2 + u^2$ therefore form an integer right triangle with $b_d > m$. A common odd prime divisor of $2uv$ and $v^2 - u^2$ would divide both u and v; and b_d is odd. Thus the triangle is primitive. The value $(Ar/d)^2 - 4d^2$ strictly decreases with d, so these triangles are distinct. Infinitely many r give distinct m with least prime factor two. This completes the proof.

4 Scope and verification

The two-triangle conclusion is already established in the OEIS comment by Aditya P Bhattad (29 April 2026). This note supplies a self-contained construction with any prescribed finite number of triangles. It is an elementary extension, not a claim to have resolved a previously open conjecture. The argument uses only infinitude of primes, elementary algebra, and coprimality. ChatGPT generated and checked this proof, including finite exact-integer tests; it has not received independent human review or formal verification.

References

[1] OEIS Foundation, *A369871*, including A. P. Bhattad's proof comment, `https://oeis.org/A369871`, consulted 11 November 2026.

[2] OEIS Foundation, *A024411*, `https://oeis.org/A024411`, consulted 11 November 2026.